

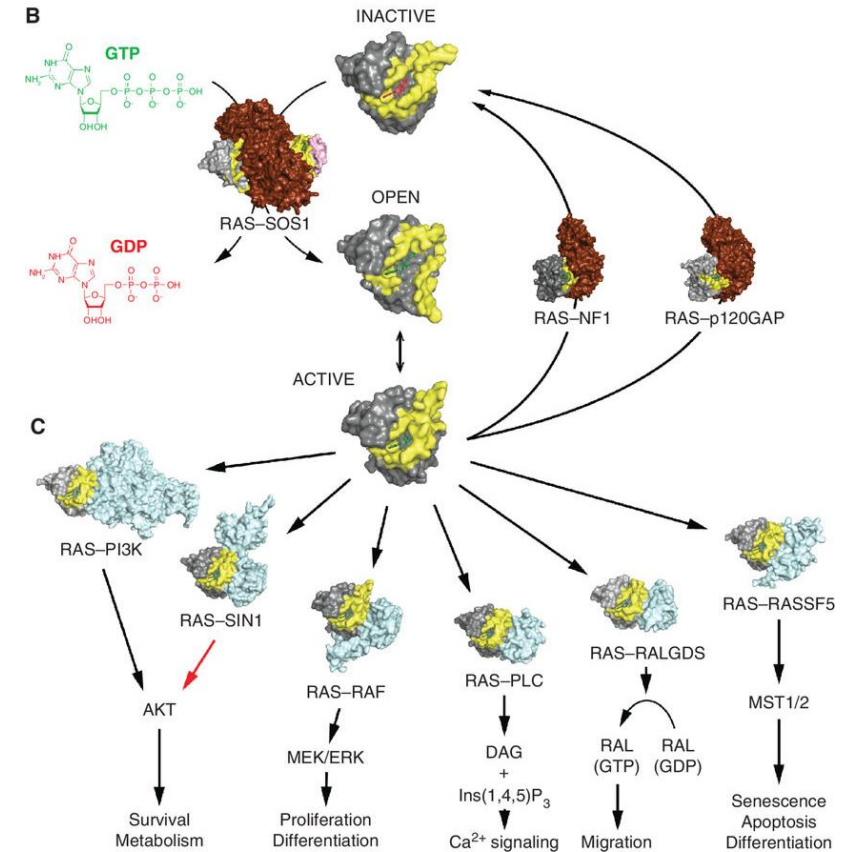
DATA UPDATE 8/28/26

Daniel Simpson



BACKGROUND

- KRas is a small GTPase responsible for regulating the MAPK signaling pathway.
- The MAPK pathway is responsible for cell growth, differentiation, and survival.
- KRas is 'active' when GTP-bound and 'inactive' when GDP-bound.
- KRas can be activated by guanine nucleotide exchange factors (GEFs) like SOS1, and deactivated by GTPase activating proteins (GAPs).



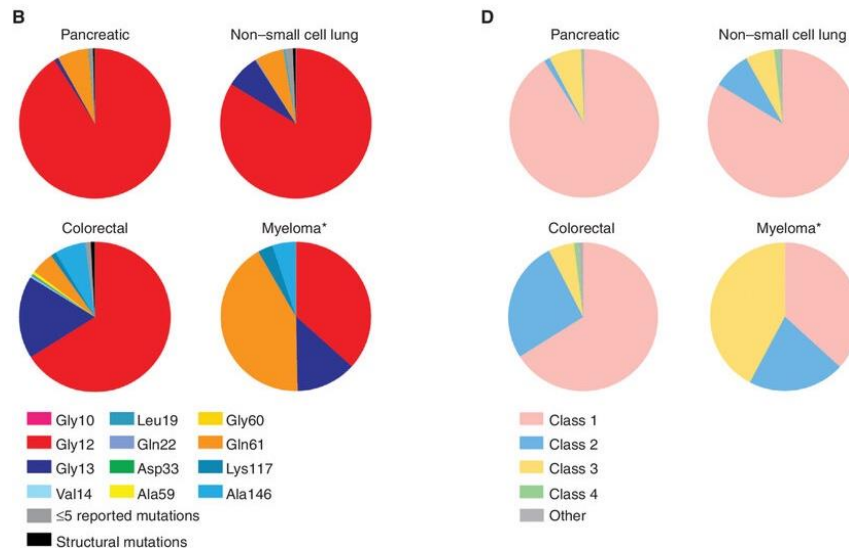
Johnson, C.W. et al., *Cancer Discovery* **2022**, 12 (4), 913–923.



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BACKGROUND CONTINUED

- Mutations in the KRas sequence can lead to the accumulation of KRas-GTP.
- This hyperactivation can occur through reduced hydrolysis (type 1), enhanced nucleotide exchange (type 2), or a combination (type 3) – and can lead to uncontrolled cell growth.
- KRas mutations are present in 10% of all human cancers.



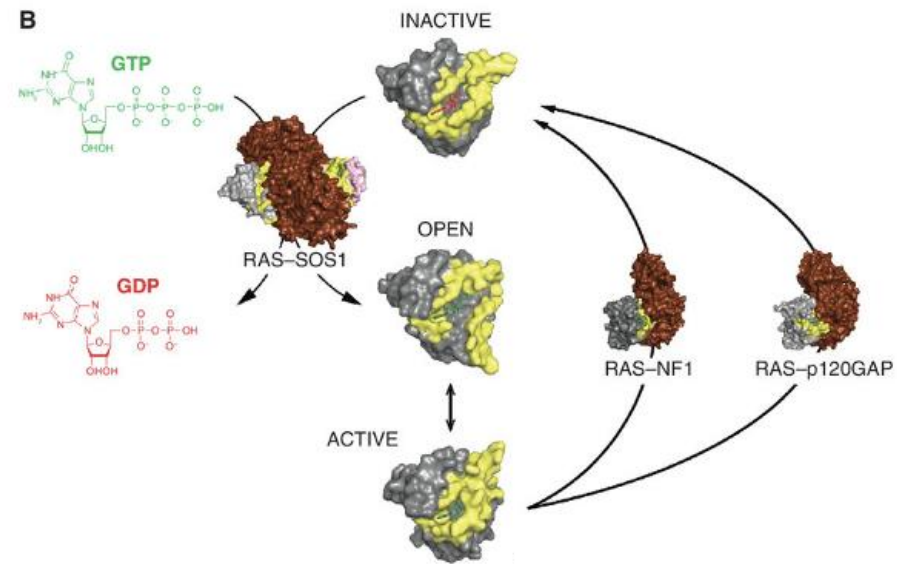
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RESEARCH QUESTIONS

1. Can we characterize the binding affinities of different KRas mutants for SOS1?
2. Can we characterize the intrinsic and GAP-catalyzed GTP hydrolysis rates for different KRas mutants?



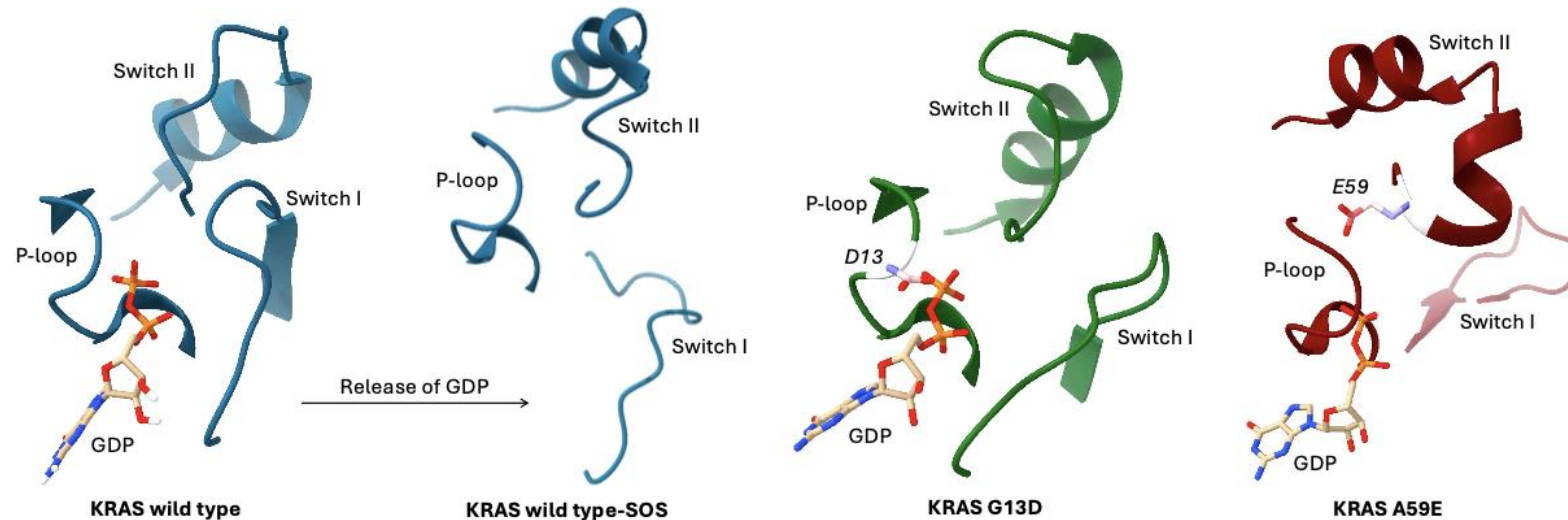
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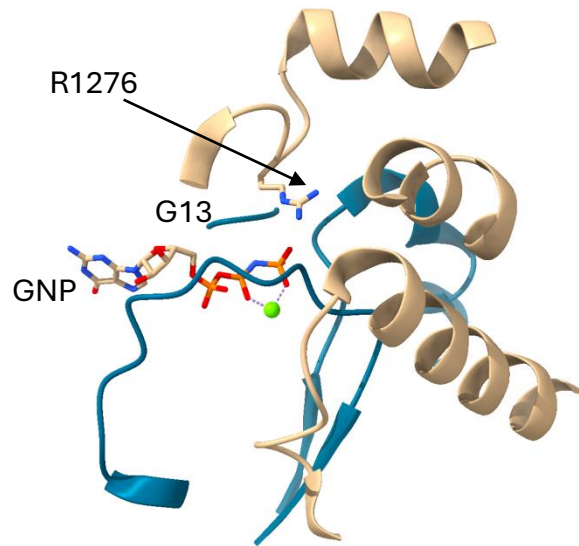
HYPOTHESIS

- Intrinsic exchange mechanism of KRas G13D mutant mocks the SOS-catalyzed exchange mechanism – so we expect a higher binding affinity of G13D for SOS.

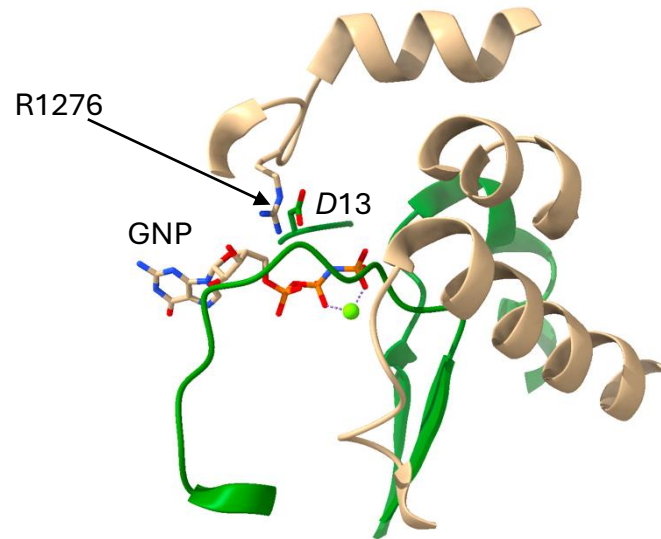


HYPOTHESIS CONTINUED

- GAP-catalyzed hydrolysis rates will be especially lower for mutants, and less so than the intrinsic rates.



KRAS WT-NF1

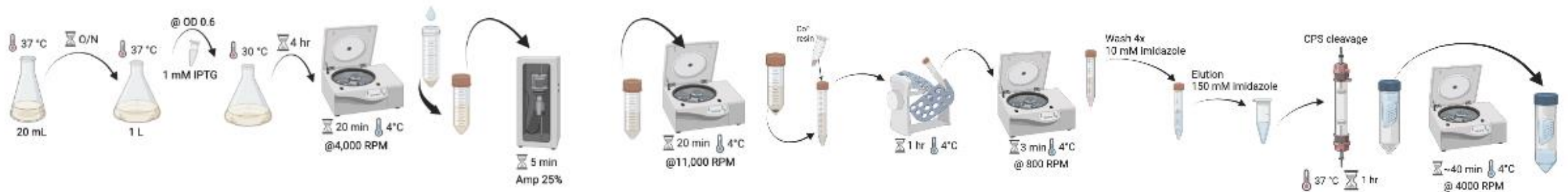


KRAS G13D-NF1



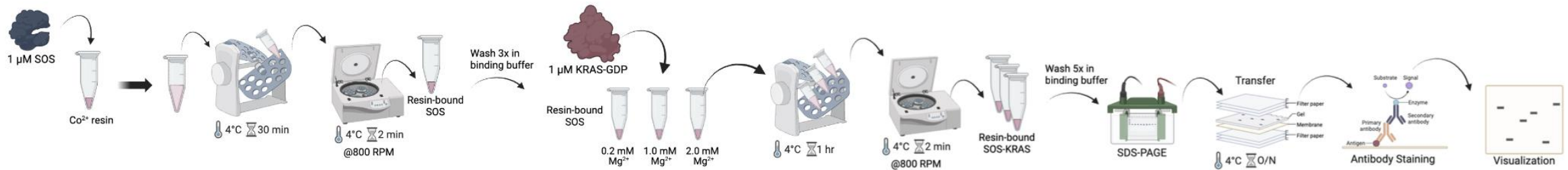
METHODS - PURIFICATION

1. Overnight culture, dilution the next morning
2. Induction with IPTG @OD 0.6, express for 4 hours.
3. Pellet, resuspension, sonication lysis
4. Incubation with resin, washing, elution
5. CPS cleavage, concentration, nucleotide* preloading



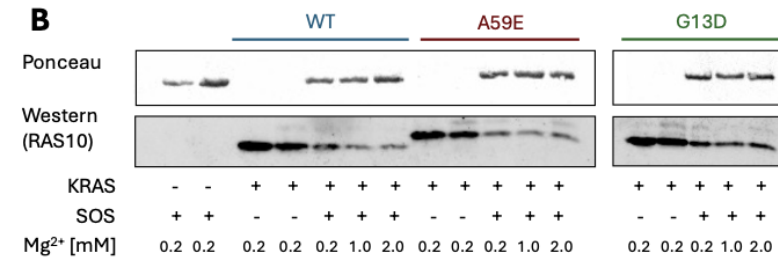
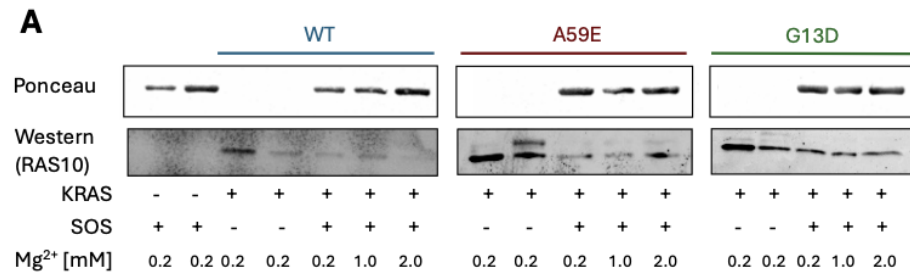
METHODS – BINDING ASSAY

1. Incubation of 6xHis-tagged SOS1 with cobalt resin.
2. Wash away unbound SOS1 (binding buffer).
3. Incubation of KRas with resin-bound SOS at varying Mg^{2+} concentrations.
4. Wash away unbound KRas.
5. Run SDS-PAGE, transfer to membrane, and incubate with antibodies.

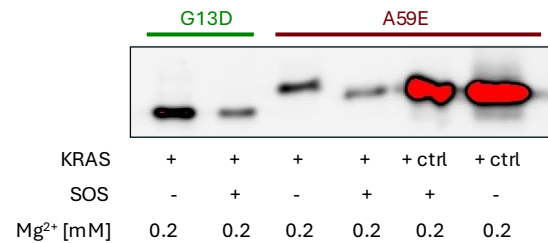


RESULTS - BINDING

- From last spring:

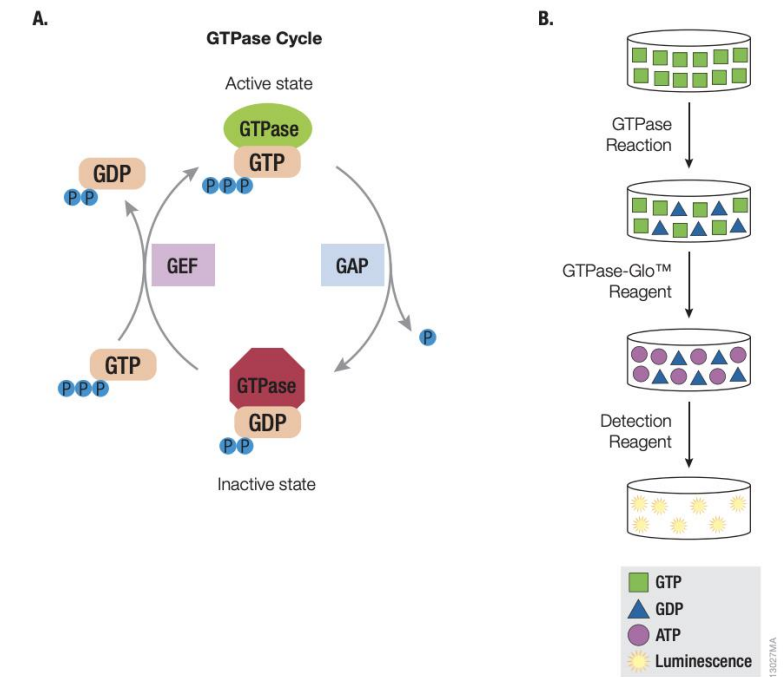


- From this summer, repeated with gentler wash steps:



METHODS - HYDROLYSIS ASSAY

1. Incubation of KRas-GTP in 2x GTP solution @ RT for 1-2 hr – with or without GAP (NF1).
2. Shaking incubation of reaction plate with GTPase-Glo reagent for 30 minutes.
3. Incubation with detection reagent for 5-10 minutes @ RT.
4. Quantification with microplate reader.



FUTURE STUDY

- Surface plasmon resonance (SPR) to determine dissociation constants of the SOS1 complexes with KRas wild type, G13D, and A59E.



CONCLUSIONS

- Binding assays showed no complex formation – or transient complex formation, although nucleotide exchange assays confirm activity of SOS.



QUESTIONS?

Name

Title

Email

Social



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